A Look at Technology in Immigrant Communities

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1. Introduction / Methodology

- Investigated impact and availability of technology on Hispanic immigrant communities
- Hypotheses drawn from conversations, survey data with students
- Found datasets for information about specific counties including computer ownership, broadband access, standardized testing data
- Explored possible correlation
- Analyzed fit (r^2 value of linear trendline)

Abstract

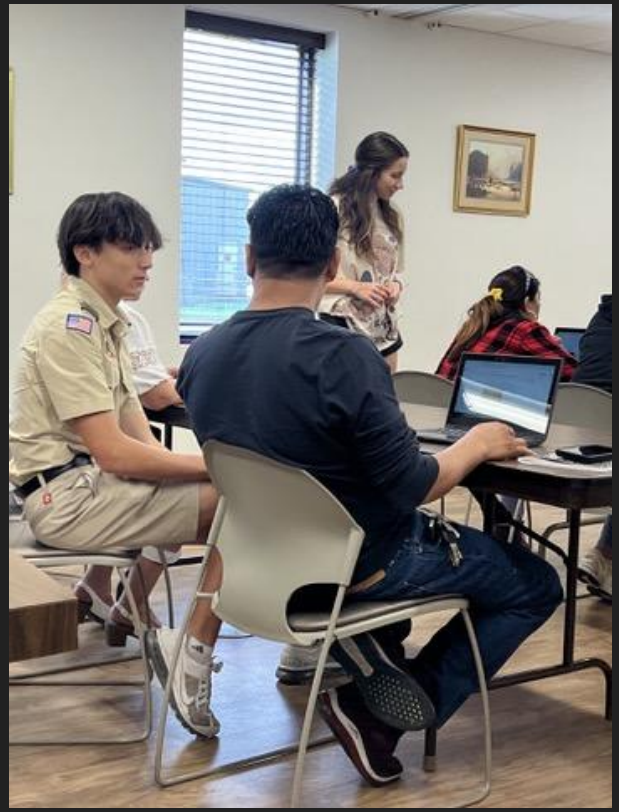
- Language achievement: internet, computer access strongly correlated, but wealth may also be a large factor
- Job acquisition: computer access & literacy significantly contribute

2. Effect of Computers on Language Learning

- Sites like Duolingo, Babbel allow easier online learning
- Accessing digital media (TV shows, music, social media) allows for better, more accessible immersion to the language
- Online classes
- Communication

2.1 Effect on Youth / Young Adults

- More successful in school assignments, working at-home
- Access numerous online resources (YouTube, Google)
- Similar effects with social media, language immersion, communication



2.2 Comparison with Testing Data

- Used python with matplotlib, numpy, pandas libraries to read, compute, and visualize the data
- Created average by multiplying # kids by average score of each school district in the county, then dividing by total population
- Graph data, use numpy “polyfit” method to add line of best fit
- 2017

```
x = x_filtered
y = y_filtered
plt.xlabel("Value (Dollars)")
plt.ylabel("% of Households to Own a Computer in the County")
plt.scatter(x, y)

# Calculate linear regression
z = np.polyfit(x, y, 1)
p = np.poly1d(z)

# Plot trendline
plt.plot(x, p(x), color="purple", linewidth=3, linestyle="--")

# Add equation of the line as annotation
equation_text = f'y = {z[0]:.6f}x + {z[1]:.6f}'
plt.annotate(equation_text, (0.05, 1.07), xycoords='axes fraction', fontsize=8)

# Calculate R-squared value
residuals = y - p(x)
SS_res = np.sum(residuals**2)
SS_tot = np.sum((y - np.mean(y))**2)
r_squared = 1 - (SS_res / SS_tot)

# Add R-squared value as annotation
r_squared_text = f'R^2 = {r_squared:.4f}'
plt.annotate(r_squared_text, (0.05, 1.02), xycoords='axes fraction', fontsize=8)

plt.show()
```

2.2 Comparison with Testing Data

```csv
"Group","District","DistName","County","CntyName","Region","RegnName","English","Math","Reading","Science","Compos","Grads_Mskd","Exnees_Mskd",
```

```csv
"At Risk",="101912","Houston ISD",="101","Harris County",="04","Houston",="16.4",="17.9",="18.1",="18.1",="17.8",="6917",="629",="9.1
"Not At Risk",="101912","Houston ISD",="101","Harris County",="04","Houston",="23.6",="23.7",="24.7",="23.5",="24.0",="3696",="1242",
```

```python
def find_index(county_name):
    for num in range(len(county_names_income)):
        if county_names_income[num] in county_name or county_name in county_names_income[num]:
            return num
    return -1

def createOrderedSet(initialSetData, initialSetNames): # creates array of data w/ the data in the correct county order
    corresponding_set = []
    for num in range(len(true_names)):
        corresponding_set.append(0) # create empty list of correct length

    for num in range(len(initialSetData)):
        county_name = initialSetNames[num]
        index = find_index(county_name)
        if (index != -1):
            corresponding_set[index] += initialSetData[num]
    return corresponding_set
```

```python
def createTestingSet(initialSetData, initialSetNames, initialPopulationSet):
    corresponding_set = []
    for num in range(len(true_names)):
        corresponding_set.append(0)
    county_name_old = ""
    total_population = 0
    index = 0
    for num in range(len(initialSetData)):
        county_name = initialSetNames[num]
        if (county_name == county_name_old and num != len(initialSetData) - 1): # if we need to add to the population, eventually get average, etc
            if (index != -1):
                total_population += int(initialPopulationSet[num])
                corresponding_set[index] += (int(initialSetData[num]) * int(initialPopulationSet[num]))
        else:
            county_name_old = county_name # adjust the county name
            print(county_name)
            if (total_population != 0):
                corresponding_set[index] = int(corresponding_set[index]) / int(total_population) # divide the old index by the population to get average score
            else:
                corresponding_set[index] = 0
            index = find_index(county_name)
            total_population = 0
            if (index != -1):
                total_population += int(initialPopulationSet[num])
                corresponding_set[index] += (int(initialSetData[num]) * int(initialPopulationSet[num]))
    return corresponding_set
```

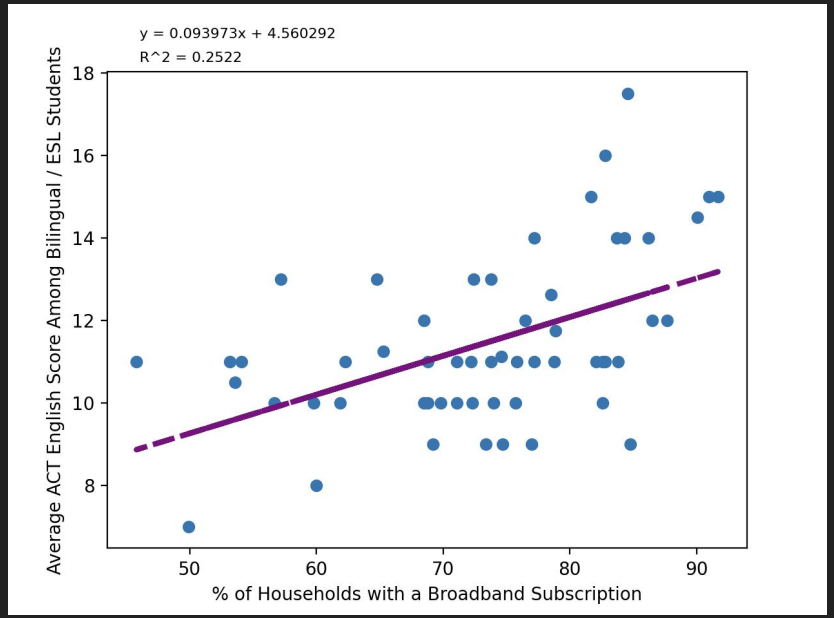
2.2 Comparison with Testing Data

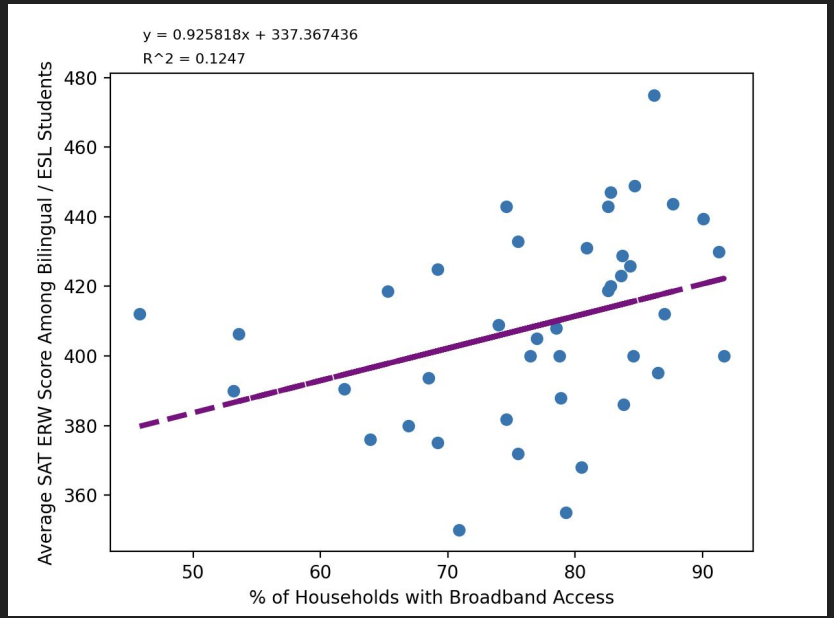
Data Excluded:

- School districts where # Bil/ESL students = 0
- Counties where # Bil/ESL students = 0
- If data for county not present in current dataset (broadband subscription, income)

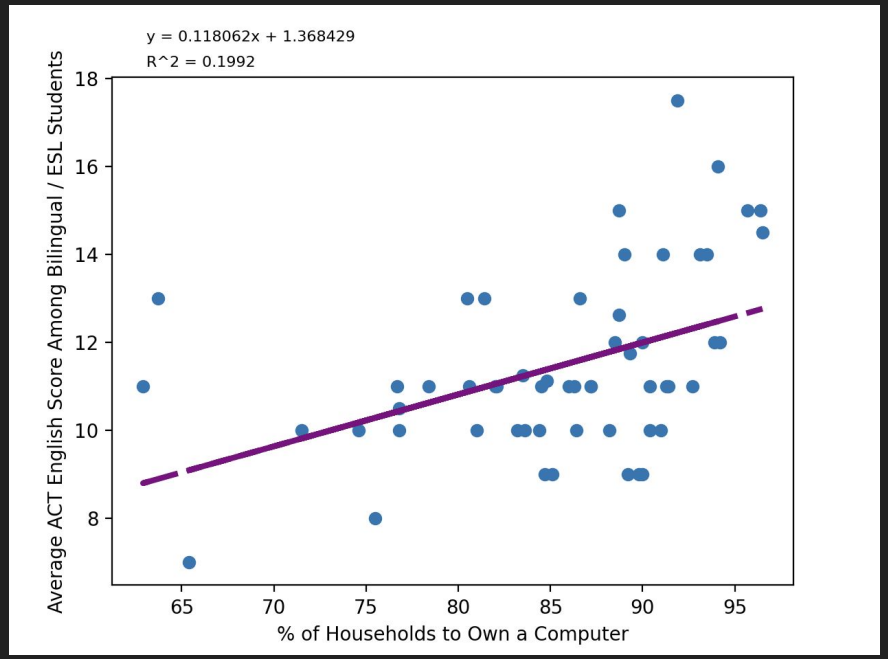
```
# Find indices where either x or y is zero
indices_to_remove = [i for i in range(len(x)) if x[i] == 0 or y[i] == 0]

# Remove corresponding indices from both x and y
x_filtered = [x_val for i, x_val in enumerate(x) if i not in indices_to_remove]
y_filtered = [y_val for i, y_val in enumerate(y) if i not in indices_to_remove]
```

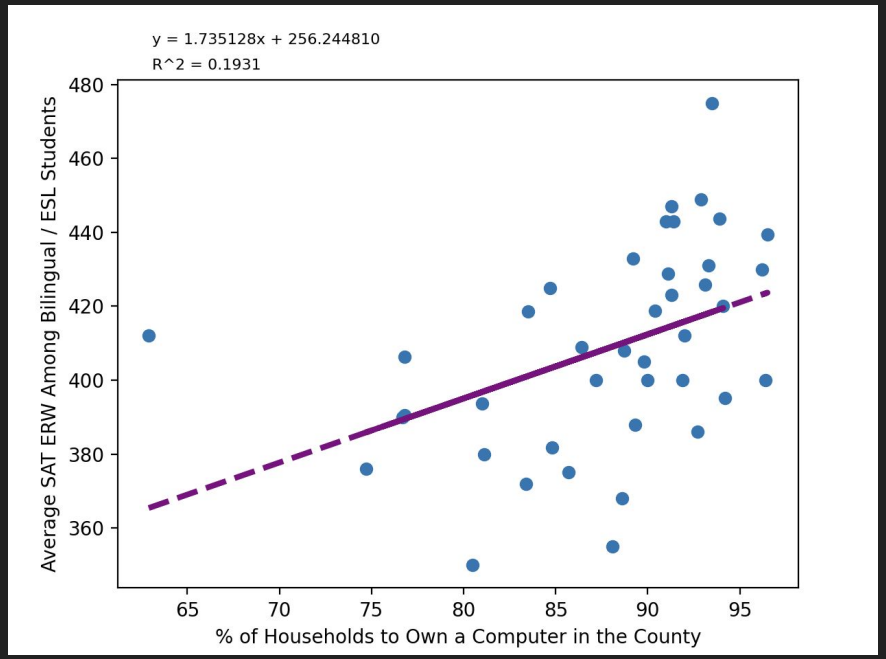
2.2 Comparison with Testing Data - Broadband





2.2 Comparison with Testing Data - Computer Ownership





2.2 Comparison with Testing Data

Comparison of fit, trend:

SAT English Reading/Writing: r^2 = trend line fit, perfect value is 1.0; m = slope

Vs. Household Computer Ownership: r^2 = .1931, m = 1.74 points / % point

Vs. Household Broadband Subscription: r^2 = .1247, m = .926 points / % point

ACT English:

Vs. Household Computer Ownership: r^2 = .1992, .118 points / % point

Vs. Household Broadband Subscription: r^2 = .2522, m = .0940 points / % point

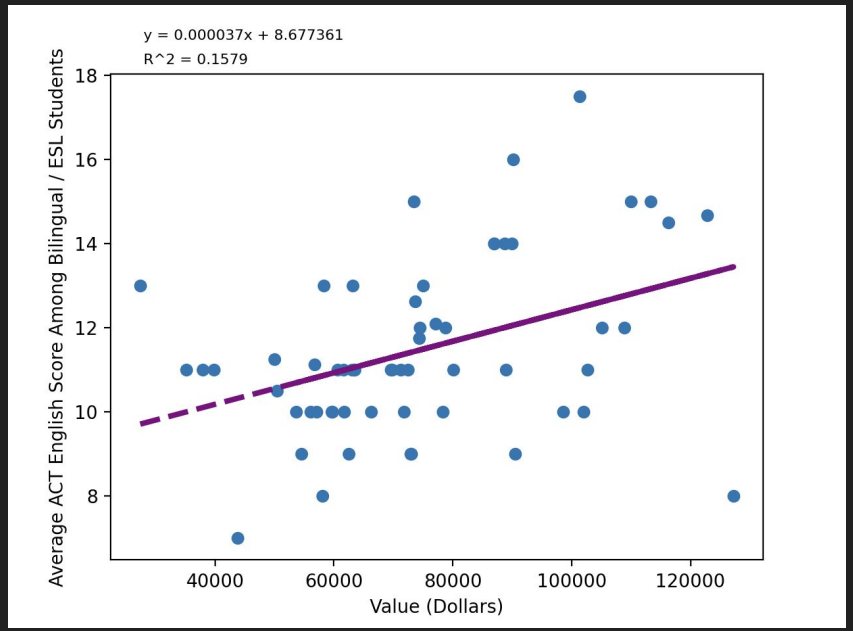
2.3 Effect of Wealth

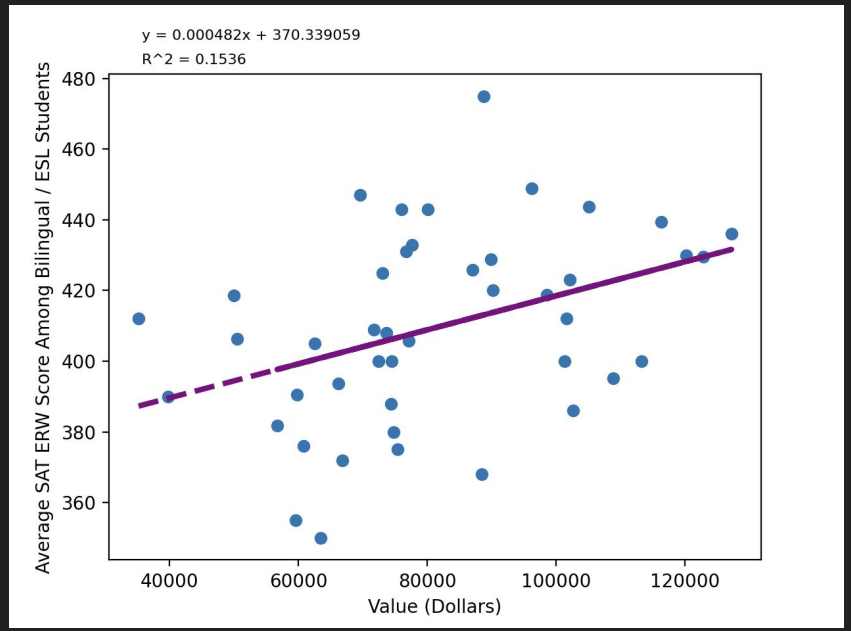
- Positively correlated with every other variable tested
- Lower r^2 value

For the next four graphs:

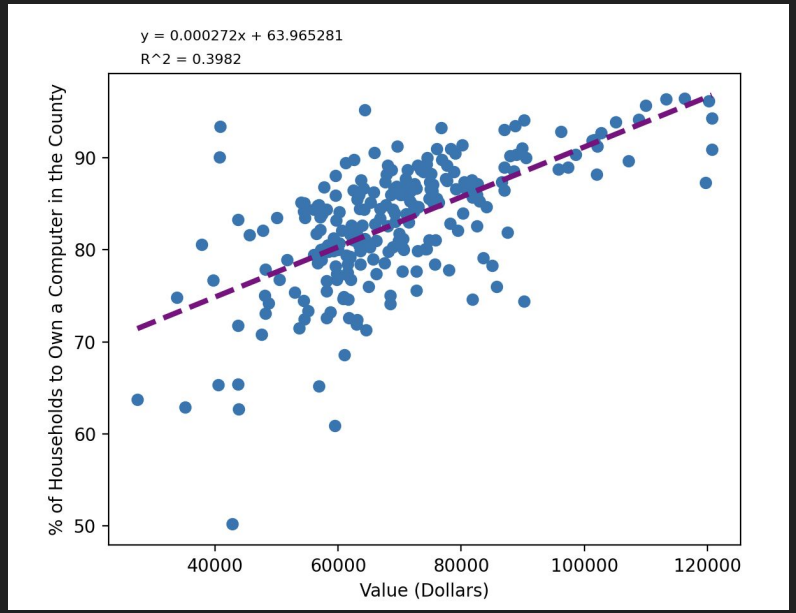
Value: Family income in USD

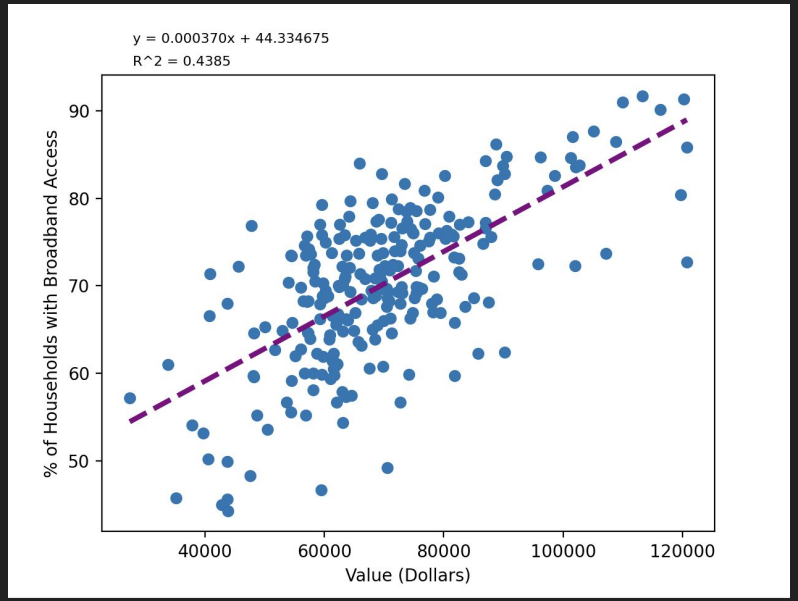
2.3 Effect of Wealth - Test Scores





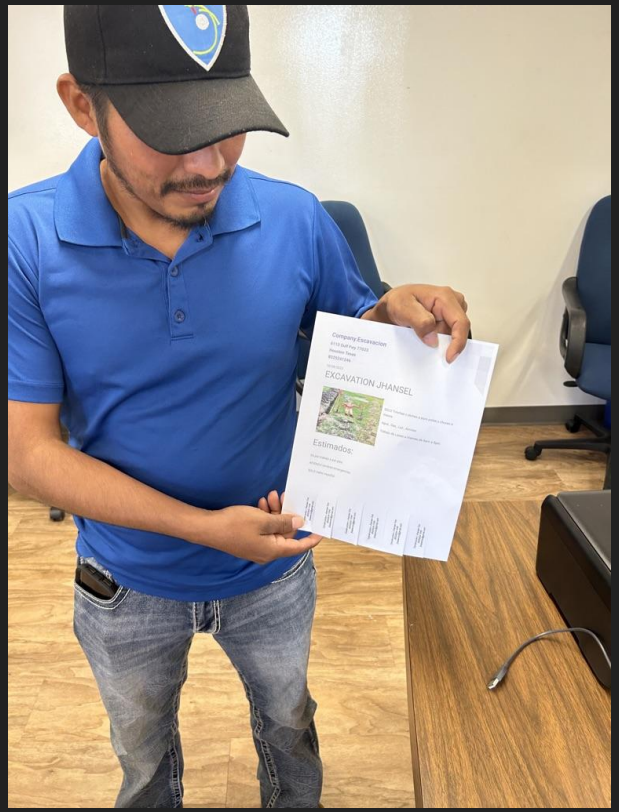
2.3 Effect of Wealth - Access to Technology





3. Job Acquisition

- All individuals surveyed acknowledged missed opportunities for work due to lack of familiarity with technology
- Modern technology (computers, internet job applications) improves chances for a position, grants individuals with new work opportunities
- Opportunities for self-promotion



3.1 Online Job Application

- 92% of employers use social media in search of talent (sites like LinkedIn)
- 40 million people look for jobs on LinkedIn every week
- 38% of all undocumented immigrants work in construction, manufacturing industries
- Another barrier

4. Necessity of Digital Literacy Education

- 1,739,000 estimated undocumented immigrants living in Texas, 52% are above the age of 34
- In 2018: just 44.7% of Mexican households contained a computer, up from 26.8% in 2009
- Compared to ~80% in the Texas overall

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